MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KU BAND, COAXIAL MAGNETRON, PULSE TYPE

This specification is approved for use by the Electronics Research and Development Command, Department of the Army and is available for use by all departments and agencies of the Department of Defense.

This specification covers the KU-band Coaxial Magnetron, Pulse Type used in Radar Set AN/MPQ-4( ). The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: The magnetron shall be coaxial type, pulsed, 15,840 - 16, 160 MHz, fixed frequency, integral magnet, forced air cooling, 70 KW minimum peak power output.

Dimensions and outline: See figure one herein

Weight: 25 lb. (max)

Coupling: Mates with Modified UG-541/U

Cooling: Forced Air (Note 9)

Mounting Position: Vertical (Note 10)

Mounting Support: Magnets (See figure one herein)

Handling: Note 8

Magnet Isolation: Note 7

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Absolute Ratings (Note 1)

a.	Parameter:	If(surge)	E_f	tk	epy	I_b	tpc	P_i	P_i	Duty Cycle
b.	Unit:	A	V	sec.	kv	a	μs	W	kW	---
c.	Maximum	70	5.3	---	26.0	14	0.3	650	360	0.002
d.	Minimum:	---	---	180	---	10	0.225	---	---	0.0015
e.	Parameter	Note 2 rrv	Body Temperature	Note 3			Note 5 Input Bushing *Temperature*		Pressurization, *Output*	Altitude
f.	Unit	kv/μs	^oC				^oC		psia	feet
g.	Maximum	180	150				200		50	10,000
h.	Minimum	130 (Notes 4,5)					---		14.7 (Note 11)	0

Test Conditions (1) (Note 6)

Parameter:	E_f(Standby)	E_f(test)	tpc	rrv	Du	I_b	VSWR
Units:	V	V	μs	kv/μs	---	mAdc	---
Value:	*4.8*	*0*	*0.25*	*145*	*0.00175*	*20*	*1.1:1*
Tolerance:	---	---	±0.025	±15	---	---	---
Notes:	3	3	5	4			11

MIL-STD-1311 Method	Requirement or Test	Conditions	AQL % (Defective)	Insp Level or Code	Symbol	Limits Min	Limits Max	Unit
	GENERAL							
---	Preproduction sample approval	required;(Note 25)	---	---	---	---	---	---
---	Marking	*(Notes 26, 27)*	---	---	---	---	---	---
---	Dimensions	(See Fig 1)	---	---	---	---	---	---
---	Packaging	(Note 28)	---	---	---	---	---	---

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QUALITY CONFORMANCE INSPECTION, PART 1 (NOTE 29)

MIL-STD-1311 Method	Requirement or Test	Conditions	AQL % (Defective)	Insp level or code	Symbol	Limits Min	Limits Max	Unit
---	Visual Mechanical Examination	(See Figure 1)	---	---	---	---	---	---
4003A	Pressurizing	Air, 45 psia Output Assembly (Note 15)	0.65	11	---	---	---	---
4289	Heater Current	E_f = 4.8 V (Standby) Tk = 270 sec.(Min.) (Note 3)	0.65	11	I_f	8	13	A
4303	Heater-Cathode Warm-up Time	E_f = 4.8 V (Standby) tk = 270 sec. (Min.) E_f = 0.0 V for test	---	---	---	---	---	---
---	Frequency	Test (1) T anode = 70 $\pm$15°C (Note 16)	---	---	F	15.840	16.160	GHz
4306	Pulse Voltage	Test (1)	---	---	epy	22	26	kv
4250	Average Power Output	Test (1)	---	---	Po	122	---	W
4308	Spectrum Measurement	Test (1)						
	R.F.Bandwidth	VSWR = 1.5:1 (Note 17)	---	---	BW	---	8	MHz
4315	Stability	Test (1) VSWR = 1.5:1 (Note 14)	---	---	MP	---	0.75	%

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MIL-STD-1311 Method	Requirement or Test	Conditions	AQL & Defective	Insp Level or Code	Symbol	Limits Min.	Limits Max.	Unit
QUALITY CONFORMANCE INSPECTION, PART 2								
1031	High Frequency Vibration	No Voltages (Note 19)	6.5	S3	---	---	---	---
4311	Frequency Pushing Factor	Test (1) VSWR = 1.5:1 max. Ib swept 16 to 20 ma (Note 21)	6.5	S3	Δ F/ ΔA	---	1.7	MHz/A
4310	Frequency Pulling Factor	Test (1) VSWR = 1.5:1 (Note 18)	6.5	S3	Δ F	---	20	MHz
QUALITY CONFORMANCE INSPECTION, PART 3, LIFE								
---	Cycling Life Test	Group D Test (1)			---	500	---	cycles
		VSWR = 1.5:1 min. cycled through λg/2 ea 15 min. of oscillation (Notes 22, 23, 24)	---	---	---	1000	---	hours
MIL-E-1: (4.6.2)	*Life-Test-End-Points*	Test (1)						
4250	Average Power		---	---	Po	90	---	W
4306	Pulse Voltage		---	---	epy	22	26	kv
---	Frequency	(Note 16)	---	---	F	15.840	16.160	GHz
4308	Spectrum Measurement RF Bandwidth	VSWR = 1.5:1 (Note 17)	---	---	BW	---	8	MHz
4315	Stability	VSWR = 1.5:1 (Note 14)	---	---	MP	---	1.0	%

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QUALITY CONFORMANCE INSPECTION, PART 3, PERIODIC CHECK (NOTES 30, 31, 32)

MIL-STD-1311 Method	Requirement or Test	Conditions	AQL % (Defective)	Insp level or code	Symbol	Limits Min	Limits Max	Unit
4028	Low Pressure	Test (1) VSWR = 1.5:1 Pressure = 522 mm/Hg (Note 13)	---	---	---	---	---	---
1026	Low Temperature	Test (1) Pre-test Cooling = 60 minutes at -65°C tk = 270 sec. max. VSWR = 1.5:1 (Note 14)	---	---	MP	---	2.0	%
1143A	Air Cooling	Test (1) T ambient = 25°C. Pin = 480 W + 10% Air Flow = 40 CFM (Note 9)	---	---	Δ T	---	60	°C
4266	Input Capacitance	Cathode Terminal to Anode	---	---	C	10	20	pf
4027	Temperature Co-efficient	Test (1) VSWR = 1.1:1 (Note 12)	---	---	Δ F/°C	---	0.75	MHz/°C
1042B	Shock	No Voltages G=15 Duration = 11 msec. (Note 20)	---	---	---	---	---	---
4315	Stability	VSWR = 1.5:1 (Note 14)	---	---	MP	---	1.0	%

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NOTES:

1. Refer to specification MIL-E-1, Section 6, "Note" covering Absolute ratings.

2. The heater surge current should not be allowed to exceed 70 amperes peak on the first cycle.

3. Prior to the application of high voltage, the cathode shall be heated to the required initial operating temperature. This is accomplished by applying 4.8 volts to the heater for 270 seconds. Upon the application of anode power, the heater voltage must be reduced to the voltage specified. The tube heater shall be protected against arcing by the use of a connector that places a minimum capacitance of 4,000 pf across the heater directly at the input terminals.

4. The rate-of-rise of voltage (rrv) shall be expressed in kv/us and is defined by the steepest tangent to the leading edge of the voltage pulse above 50% amplitude. Any capacitance used in the viewing system shall not exceed 6.0 pf.

5. The characteristics of the applied pulse must be those that result in proper starting and oscillation. The rate of pulse voltage rise, the percentage of pulse voltage ripple, and the rate of pulse voltage fall are among the more important considerations. Negative post pulse voltage oscillation should be minimized to avoid leakage current and noise. Values of voltage and current fall time, spike, ripple, inverse voltage and post voltage oscillations should be minimized.

6. The pulse characteristics for Test Condition (1) test shall be as follows:

	Min.	Max.	Unit
tpc	0.225	0.275	μsec.
tfc	---	0.15	μsec.
rrv	130	160	kv/usec.
Ripple (current)	---	±10	%
Inverse Voltage	---	20	%
Post Pulse Forward Voltage	---	5	%

7. In handling, transportation, and mounting the tube, care shall be exercised to prevent demagnetization. A minimum distance of 8 inches shall be maintained between the tube magnets and any other ferromagnetic material. Other energized magnets must not be permitted to be closer than 12 inches from the tube magnet.

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8. Care should be exercised during handling, storage, and use of the tube to prevent greater shock or vibration than that for which the tube is designed.

9. Anode cooling-fin inlet duct for supplying air to the tube cooling system must be such as to insure delivery of air to the entire cross-section of the inlet with minimum side leakage.

10. The tube shall be mounted with the cathode vertical to within 15° deviation.

11. The waveguide must be pressurized at 14.7 psia minimum to prevent breakdown under matched conditions (VSWR = 1.1:1 maximum). Under mismatch conditions (VSWR = 1.5:1 maximum) the waveguide must be pressurized at 35 psia minimum. The gas used in pressurization shall provide insulating properties equal to or better than dry air at the pressure specified.

12. Temperature shall be measured at point shown on outline drawing. Temperature shall be measured after thermal equilibrium has been reached.

13. Under specified pressurization and with the load phase varied through $\frac{1}{2}\ \lambda_g$ minimum, there shall be no evidence of breakdown at the input assembly.

14. Stability shall be measured in terms of the average number of output pulses missing, expressed as a percentage of the number of input pulses applied during the period of observation. The missing pulses (MP), due to any cause, are considered to be missing if the energy level, with $\pm 1\%$ frequency range of the normal operating frequency, is less than 70% of the energy level of a normal pulse. The percentage of missing pulses shall not exceed the amount specified when measured during the last three minutes of a test interval not exceeding six minutes. The load shall be adjusted to that phase producing maximum instability.

15. The seal formed by clamping the output flange against a suitable test fixture shall be hermetically tight for one minute minimum with air pressure as specified applied internally to the test fixture (see outline drawing).

16. The frequency shall be within the limits specified when measured under the stipulated temperature conditions and in accordance with paragraph 4.9(h) of MIL-STD-1311.

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17. The specification limits shall not be exceeded at the point of maximum degradation determined by observation while varying a mismatch of 1.5:1 through 180 degrees.

18. The frequency deviation (maximum frequency minus minimum frequency) shall not exceed the limits specified herein when a VSWR of 1.5:1 minimum, introduced into the load at a distance of approximately 0.25 meter from the tube flange, is varied through all phases. This test shall be made in a manner such that thermal effects do not contribute appreciable error.

19. The tube shall be mounted in a resonance-free jig and vibrated with sinusoidal excitation at the specified level. Upon completion of vibration test, the tube shall meet the requirements for r.f. spectrum and stability per Quality Conformance Inspection, Part 1, of this specification.

20. The magnetron shall be mounted on a test plate and subject to five half-sine shock pulses in each of three mutually perpendicular axes parallel to the reference planes shown in outline drawing. After completion of shock tests, each tube shall meet the requirements for r.f. spectrum and stability per Quality Conformance Inspection, Part 1, of this specification.

21. The pushing factor shall be determined for the specified current range. The test shall be conducted to exclude the effects of thermal drift and frequency instability not due to pushing.

22. Life test shall be conducted in accordance with the following cycle, which defines one complete life test cycle.

Test Condition	Ib (mAdc)	E_f (Volts)	Duration (Minutes)
Standby	0	4.8	4.5
1	18 (with 10% blanking, Note 23)	0	120.0
Off	0	0	25.5

23. The anode pulse applied to the magnetron during the r.f. on -time of the life cycle shall be alternately on for 27,000 microseconds and off for 3,000 to represent the scanner blanking time of the AN/MPQ-4( ) radar set . The initial post blanking peak anode current shall not exceed 14.0 amperes.

24. The number of samples to be life tested shall be in accordance with MIL-E-1 or as specified in the contract.

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25. Preproduction Sample Approval in lieu of Qualification - Approval of production sample, based upon preproduction sample inspection, shall replace Qualification requirements of Specification MIL-E-1. Preproduction sample tests, testing, sampling, and approval criteria shall correspond directly with all pertinent requirements in the specification as normally applicable to qualification of product.

    a. Preproduction tests will be performed after award of contract, at the manufacturer's plant, unless otherwise specified in the bid request or contract. Approval of the preproduction sample is required prior to initial delivery of units on the contract. The fabrication of production units, prior to approval of the preproduction sample, shall be at the manufacturer's risk.

    b. Unless otherwise specified in the bid request or contract, the specimens shall become the property of the government and, upon approval, shall serve as a reference standard for the contracting officer and government inspector.

26. Tube-designation marking shall be in accordance with Specification MIL-E-1

27. Tube identification and tube serial number shall be marked in the most practical and most readily discernible location on the tube, relative to its normal mounted position (See Note 10, above).

28. Each tube shall, as a minimum protection against moisture and moisture vapor be sealed within a moisture-vaporproof barrier bag, or equivalent, normally acceptable under packaging requirements of Specification MIL-E-75. Also each electron tube shall be furnished with a dust cover for protection of the output connector.

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29. The AQL for the combined defectives for attributes in Quality Conformance Inspection, Part 1, exclusive of "Visual-Mechanical examination", shall be 1.0%. A tube having more than one defect shall be counted as one defective.

30. For periodic tests, sample tubes are to be selected at random intervals so that during only twelve-month period a minimum of two tubes shall have been successfully tested. In the event of failure of one or more tubes in a sample, these tests are to become a part of Quality Conformance Inspection, part 2, until two successive lots have been successfully completed on these tests. After two successive lots have thus been sucessfully completed, the periodic tests shall then be eligible to be resumed on a twelve-month basis.

31. After subjection to all Periodic-check tests, the sample tube(s) shall meet the requirements of all tests in QCI, Part 1 herein.

32. Although all tests in Part 3, Periodic check, herein may be performed in any sequence, the sequence recommended is as follows:

Air cooling
Input capacitance
Frequency
Temperature coefficient
Low pressure
Low-temperature operation
Shock

However, at option of the manufacturer, the tube sample(s) may be subjected to the Part 3, Periodic Check tests herein in accordance with the following grouping plan (see c, below):

a. The tube sample(s) shall be subjected first to all tests in Group 1, then to all tests in Group 2, and in the test sequence shown within each group. The requirement of Note 31 above shall be effective for performance immediately after the last test in each group.

b.

Group 1	Group 2
Air cooling	Low-temperature operation
Input capacitance	Shock
Low pressure	
Frequency	
Temperature Coefficient	

c. The above grouping plan is provided to relieve the need to repeat pre- and post-test end-point measurements (all tests of QCI-1 herein) normally pertinent to individual tests such as Shock and Humidity.

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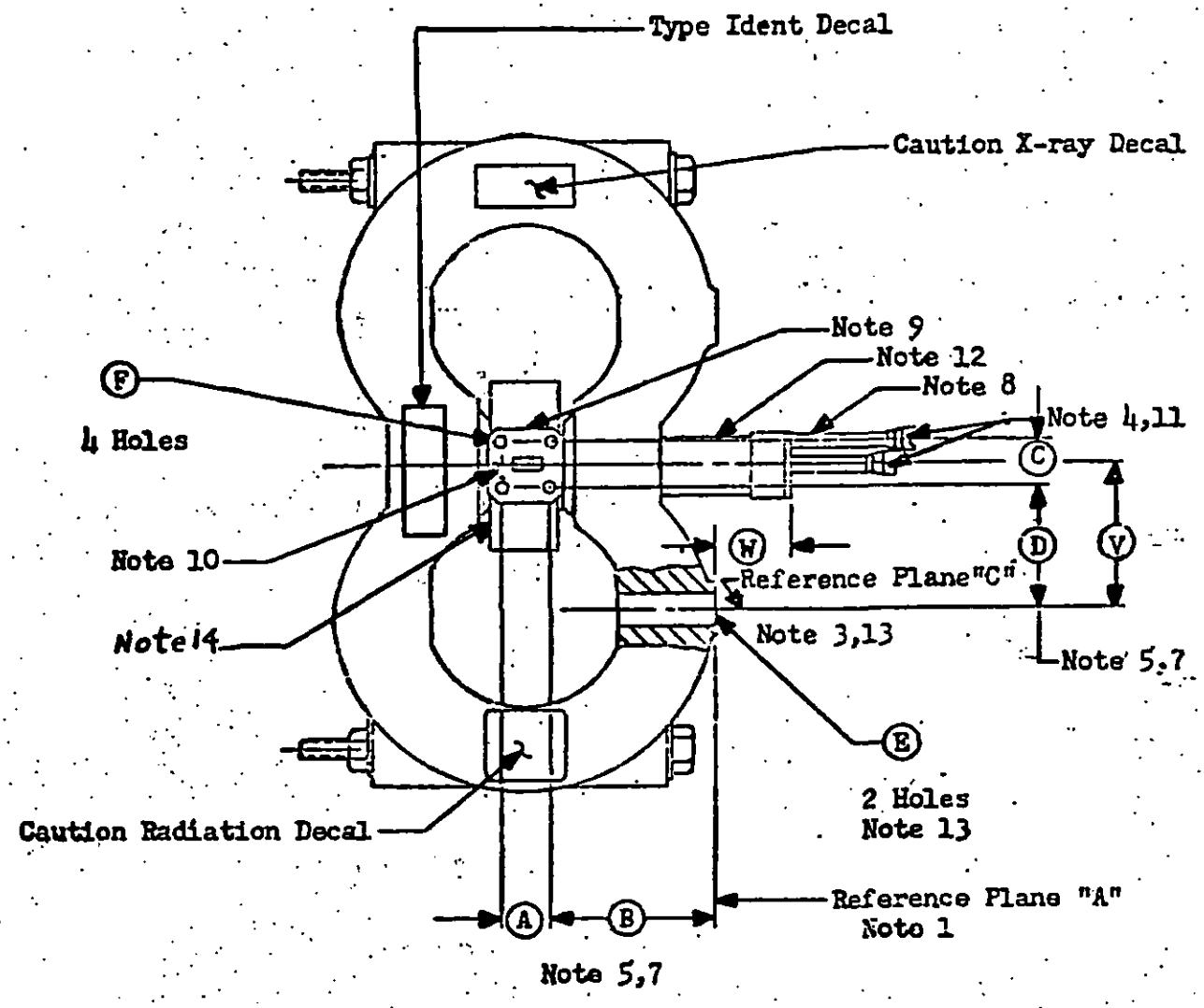


Figure 1

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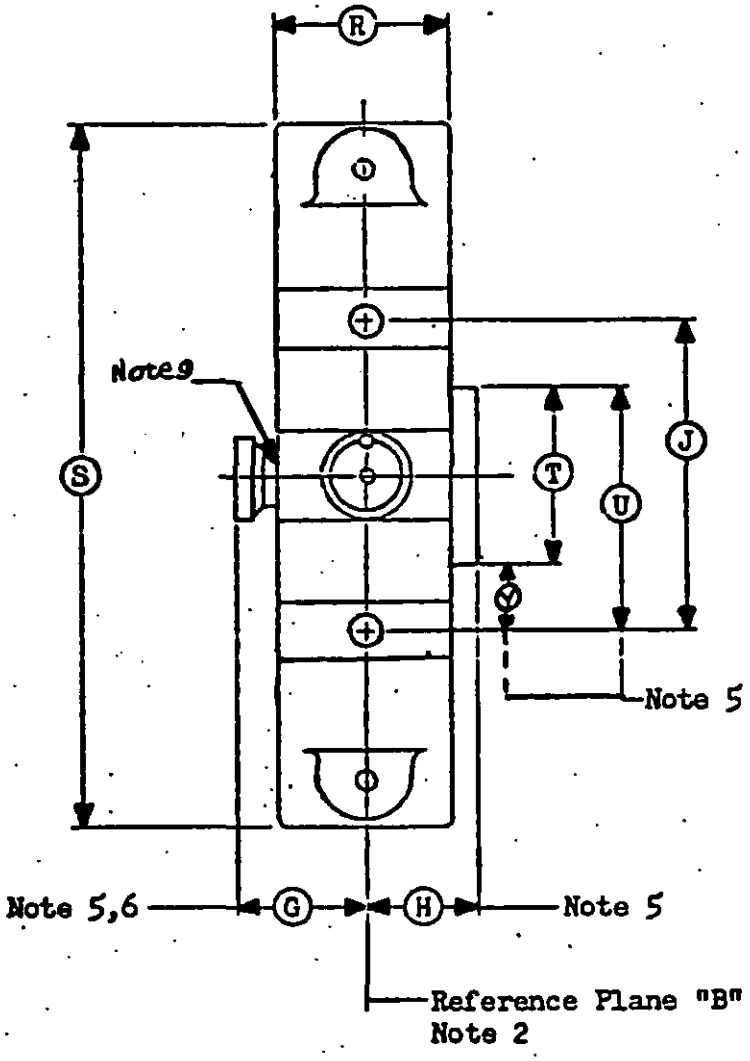


Figure 1 (cont'd)

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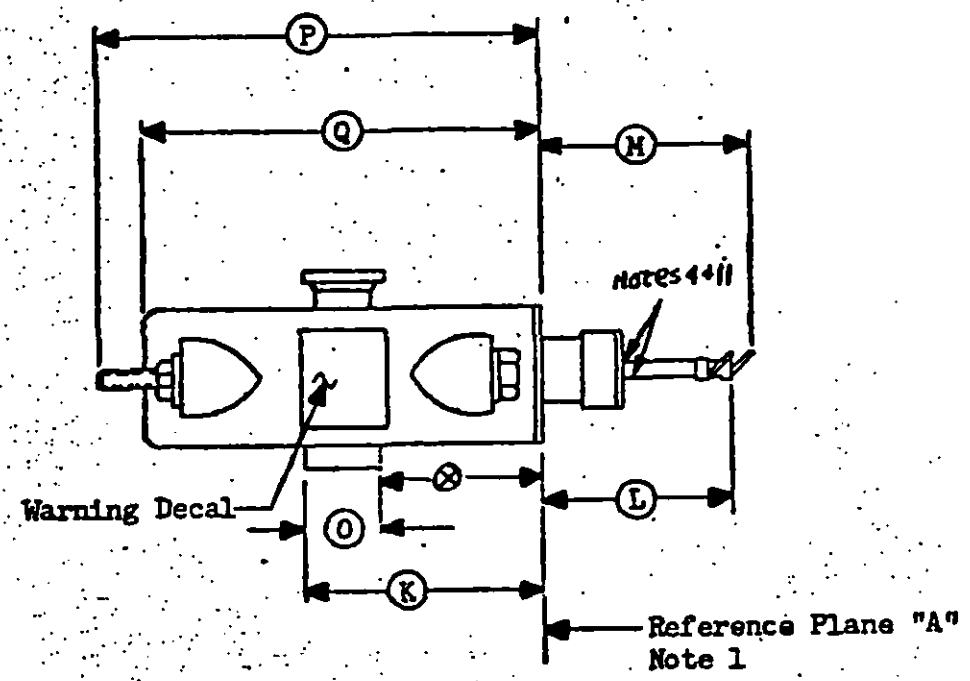


Figure 1 (cont'd)

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REF	Dimension (inches)		
	Max.	Min.	
A	1.004	.984	*
B	2.639	2.499	
C	.966	.946	*
D	2.217	2.077	
E	.531	.469	*
F	6-32 NC-2B		*
G	1.970	1.870	
H	2.000		*
J	5.322	5.178	*
K	3.780		*
L	8.750	8.250	
M	9.000	8.500	
O	1.135	.985	*
P	7.250		*
Q	6.500		*
R	3.250		*
S	12.500		*
T	3.070	2.930	*
U	4.281		*
V	2.750	2.500	*
W	3.312		*
X		2.345	*
Y		.963	*

*Indicates Design Reference Dimensions

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NOTES FOR 5434 OUTLINE DRAWING:

1. Reference plane "A" is defined as a plane passing along the finished surface of the magnet.

2. Reference plane "B" is defined as a plane passing through the centers of the mounting holes, and perpendicular to reference plane "A".

3. Reference plane "C" is defined as a plane passing through the center of the mounting hole, and mutually perpendicular to reference planes "A" and "B".

4. Wire: Insulated, No. 11 AWG; 7 x 6 x 30 AWG, or equal.

   Lugs: Burndy spade lug YAVIO-T21F, or equal. One lug painted brown to identify it as common heater-cathode connection, outer terminal. Paint shall not be applied to that portion of spade lug used for electrical connection.

5. Tolerance includes angular as well as lateral deviation(s).

6. Surface of output flange to be parallel to reference plane "B" within 0.020. Condition of this surface to be suitable for a hermetically tight seal.

7. Tube-to-tube variation shall be within the deiensional tolerances, but hole-to-hole variation on any single tube shall not vary more than 0.030 in. within the dimensional tolerances.

8. Minimum distance from high voltages to ground should be 1.5 inches through air at normal atmospheric pressure.

9. Point at which temperature shall be measured on anode body.

10. Mates with modified UG-541/U flange.

11. High voltage terminals; center terminal = heater; outer terminal = heater-cathode. Wire connections (4 above) to the high voltage terminals shall be mechanically and electrically firm to prevent loosening during handling, installation and operation.

12. Point at which temperature shall be measured on cathode bushing.

13. Mounting holes: entire support and fastening to be effective therein.

14. Anode body must be grounded effectively through tube mounting holes.

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